

EMF & FARADAY'S LAW

Transformer EMF *stationary loop, B changes*

$$V_{\text{emf}} = - \int_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{s}$$

Motional EMF *loop moves, B static*

$$V_{\text{emf}} = \oint (\mathbf{v} \times \mathbf{B}) \cdot d\mathbf{l}$$

General (flux form) *always works*

$$V_{\text{emf}} = - \frac{d\Phi}{dt} \quad \Phi = \int_S \mathbf{B} \cdot d\mathbf{s}$$

N-turn coil *transformer, N turns*

$$V_{\text{emf}} = -N \frac{d\Phi}{dt} = - \frac{d\lambda}{dt} \quad \lambda = N\Phi$$

Uniform B shortcut *B same everywhere on loop*

$$V_{\text{emf}} = - \frac{\partial B}{\partial t} \cdot A$$

Long wire + rect loop *wire near loop, r: a → b, height c*

$$V_{\text{emf}} = - \int_0^c \int_a^b \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{s} = - \int_0^c \int_a^b \frac{\partial}{\partial t} \left(\frac{\mu_0 I}{2\pi r} \right) dr dz$$

MAXWELL'S EQUATIONS

Name	Differential form
Faraday	$\nabla \times \mathbf{E} = -\partial \mathbf{B} / \partial t$
Ampère	$\nabla \times \mathbf{H} = \mathbf{J} + \partial \mathbf{D} / \partial t$
Gauss (E)	$\nabla \cdot \mathbf{D} = \rho_v$
Gauss (B)	$\nabla \cdot \mathbf{B} = 0$

$\mathbf{D} = \epsilon \mathbf{E}$; $\mathbf{B} = \mu \mathbf{H}$; *source-free: $\rho_v = 0$, $\mathbf{J} = 0$*

SURFACE ELEMENT $d\mathbf{s}$

$d\mathbf{s} = \hat{n} da$ (RHR gives $\hat{n}$: curl fingers in loop direction, thumb = $\hat{n}$)

Loop plane	$d\mathbf{s}$
x-y	$\hat{z} dx dy$ or $\hat{z} r dr d\phi$
y-z	$\hat{x} dy dz$
x-z	$\hat{y} dx dz$
coplanar / toroid	$\hat{\phi} dr dz$

Uniform B: $\iint da = A$ (just the loop area)

DOT PRODUCTS $\hat{a} \cdot \hat{b}$

Same direction = 1; perpendicular = 0

	$\hat{x}$	$\hat{y}$	$\hat{z}$	$\hat{r}$	$\hat{\phi}$
$\hat{x}$	1	0	0	0	0
$\hat{y}$	0	1	0	0	0
$\hat{z}$	0	0	1	0	0
$\hat{\phi}$	0	0	0	0	1

CROSS PRODUCT $\mathbf{A} \times \mathbf{B}$ (determinant)

$$\mathbf{A} \times \mathbf{B} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix} = (A_y B_z - A_z B_y) \hat{x} - (A_x B_z - A_z B_x) \hat{y} + (A_x B_y - A_y B_x) \hat{z}$$

Also used for curl:

$$\nabla \times \mathbf{E} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \partial_x & \partial_y & \partial_z \\ E_x & E_y & E_z \end{vmatrix}$$

Plane wave ($\partial_x = \partial_y = 0$): only ∂_z row survives.

COAXIAL CABLE — AC RESISTANCE

$$R' = \frac{R_s}{2\pi} \left(\frac{1}{a} + \frac{1}{b} \right) \quad (\Omega/\text{m})$$

a : inner radius; b : outer radius; $R_s = \sqrt{\pi f \mu / \sigma}$

B FIELDS

Infinite wire *single wire carrying I, N = 1*

$$\mathbf{B} = \hat{\phi} \frac{\mu_0 I}{2\pi r}$$

Inside toroid *coil of N turns carrying I*

$$\mathbf{B} = \hat{\phi} \frac{\mu N I}{2\pi r} \quad \mu = \mu_r \mu_0$$

N in B only if the source (current-carrying thing) is a multi-turn coil. Single wire $\Rightarrow N = 1$, drop it. N in $V_{\text{emf}} = -N d\Phi/dt$ is the receiver coil turns.

$\partial B / \partial t$ TABLE

$B(t)$	$\partial B / \partial t$
$B_0 \sin(\omega t)$	$B_0 \omega \cos(\omega t)$
$B_0 \cos(\omega t)$	$-B_0 \omega \sin(\omega t)$
$B_0 e^{-\alpha t}$	$-\alpha B_0 e^{-\alpha t}$
B_0 (const)	0

SIGN → CURRENT DIRECTION

V_{emf}	Inside source	I dir.
< 0	− → + terminal	$-\hat{\phi}$ (CW)
> 0	+ → − terminal	$+\hat{\phi}$ (CCW)
= 0	—	none

Lenz: induced I always opposes the change in Φ .

ln INTEGRAL

$B \propto 1/r$, loop spans $r = a$ to $r = b$

$$\int_a^b \frac{dr}{r} = \ln \frac{b}{a} = \ln b - \ln a$$

$\ln \frac{b}{a} = \ln b - \ln a$
 $\ln 2 = 0.693 \quad \ln 3 = 1.099 \quad \ln 1.2 = 0.182$

ROTATING LOOP — GENERATOR

$\Phi = AB_0 \cos(\omega t + \phi_0)$ $V_{\text{emf}} = AB_0 \omega \sin(\omega t + \phi_0)$ $I_{\text{amp}} = \frac{AB_0 \omega}{R} \quad \omega = 2\pi f = 2\pi \frac{\text{rpm}}{60}$ Common loop areas	
Shape	Area A
Square (side a)	a^2
Rectangle ($l \times w$)	lw
Circle (radius r)	πr^2
Equil. triangle (side a)	$\frac{\sqrt{3}}{4} a^2$
Any triangle	$\frac{1}{2} bh$

SI PREFIXES

Prefix	Sym	Value	Prefix	Sym	Value
pico	p	10^{-12}	kilo	k	10^3
nano	n	10^{-9}	mega	M	10^6
micro	μ	10^{-6}	giga	G	10^9
milli	m	10^{-3}	tera	T	10^{12}

j IDENTITIES

Expression	Result	Expression	Result
j^2	−1	$j \times (-j)$	1
$1/j$	−j	$-1/j$	j
j^3	−j	j^4	1
$e^{-j\pi/2}$	−j	$e^{j\pi/2}$	j
$e^{j\pi}$	−1	$e^{j2\pi}$	1

DISPLACEMENT & CONDUCTION CURRENT

Conduction *resistive, $\sigma \neq 0$*

$$E = \frac{V(t)}{d} \quad J_c = \sigma E \quad I_c = J_c \cdot A = \frac{\sigma A}{d} V(t)$$
$$I_c = \frac{\sigma A}{d} V(t) \iff I = \frac{V}{R} \Rightarrow R = \frac{d}{\sigma A}$$

$\underbrace{\hspace{1cm}}_{1/R}$

Current proportional to V directly $\rightarrow$ behaves like a resistor.

Displacement *capacitive, changing E*

$$J_d = \epsilon \frac{\partial E}{\partial t} \quad I_d = J_d \cdot A = \frac{\epsilon_r \epsilon_0 A}{d} \frac{dV}{dt}$$
$$I_d = \underbrace{\frac{\epsilon_r \epsilon_0 A}{d}}_C \frac{dV}{dt} \iff I = C \frac{dV}{dt} \Rightarrow C = \frac{\epsilon_r \epsilon_0 A}{d}$$

Current proportional to dV/dt $\rightarrow$ behaves like a capacitor.

Lossy cap circuit *both σ and ϵ_r present*

$$R \parallel C \text{ across } V(t): \quad I = I_c + I_d = \frac{V}{R} + C \frac{dV}{dt}$$

CHARGE-CURRENT CONTINUITY

$$\nabla \cdot \mathbf{J} = - \frac{\partial \rho_v}{\partial t}$$

Conductor (steady): $\nabla \cdot \mathbf{J} = 0 \Rightarrow \oint_S \mathbf{J} \cdot d\mathbf{s} = 0$ (KCL)

CHARGE RELAXATION

Decay *charge injected at $t = 0$*

$$\rho_v(t) = \rho_{v0} e^{-t/\tau_r} \quad \tau_r = \frac{\epsilon_r \epsilon_0}{\sigma}$$

Find σ from decay data

$$\sigma = \frac{-\ln(\rho_v/\rho_{v0}) \cdot \epsilon_r \epsilon_0}{t_1}$$

Larger τ_r = slower dissipation = better insulator.

PHASOR METHOD (E → H)

- Key relation** $\mathbf{E}(z, t) = \text{Re}\{\tilde{\mathbf{E}}(z) e^{j\omega t}\}$
- Euler split** $e^{j(\omega t - kz)} = e^{j\omega t} \cdot e^{-jkz}$ strip $e^{j\omega t}$, keep e^{-jkz}
- Time → phasor conversions**
 $\cos(\omega t - kz) = \text{Re}\{e^{j(\omega t - kz)}\} = \text{Re}\{e^{j\omega t} \cdot e^{-jkz}\} \rightarrow e^{-jkz}$
 $\sin(\omega t - kz) = \cos(\omega t - kz - \frac{\pi}{2}) = \text{Re}\{e^{j(\omega t - kz - \pi/2)}\} = \text{Re}\{e^{-j\pi/2} \cdot e^{j\omega t} \cdot e^{-jkz}\} \rightarrow -j e^{-jkz}$
- Faraday in phasor domain** (plane wave: only $\partial/\partial z$ survives)
 $\tilde{\mathbf{H}} = - \frac{1}{j\omega\mu} \nabla \times \tilde{\mathbf{E}}$
- Phasor → time domain** $\mathbf{H}(z, t) = \text{Re}\{\tilde{\mathbf{H}} e^{j\omega t}\}$
 $\text{Re}\{e^{j(\omega t - kz)}\} = \cos(\omega t - kz)$
 $\text{Re}\{-j e^{j(\omega t - kz)}\} = \sin(\omega t - kz)$

UNIT CIRCLE

Angle	Deg	cos	sin	Q
0	0	1	0	—
$\pi/6$	30	$\sqrt{3}/2$	1/2	1
$\pi/4$	45	$\sqrt{2}/2$	$\sqrt{2}/2$	1
$\pi/3$	60	1/2	$\sqrt{3}/2$	1
$\pi/2$	90	0	1	—
$2\pi/3$	120	−1/2	$\sqrt{3}/2$	2
$3\pi/4$	135	− $\sqrt{2}/2$	$\sqrt{2}/2$	2
$5\pi/6$	150	− $\sqrt{3}/2$	1/2	2
π	180	−1	0	—
$7\pi/6$	210	− $\sqrt{3}/2$	−1/2	3
$5\pi/4$	225	− $\sqrt{2}/2$	− $\sqrt{2}/2$	3
$4\pi/3$	240	−1/2	− $\sqrt{3}/2$	3
$3\pi/2$	270	0	−1	—
$5\pi/3$	300	1/2	− $\sqrt{3}/2$	4
$7\pi/4$	315	$\sqrt{2}/2$	− $\sqrt{2}/2$	4
$11\pi/6$	330	$\sqrt{3}/2$	−1/2	4
2π	360	1	0	—

FIELDS

$\mathbf{E}$ — electric field (V/m)
 $\mathbf{D} = \epsilon \mathbf{E}$ — elec. flux density (C/m^2)
 $\mathbf{B}$ — mag. flux density (T)
 $\mathbf{H}$ — mag. field intensity (A/m)
 ρ_v — charge density (C/m^3)
 ρ_{v0} — initial charge density
 $\Phi = \int_S \mathbf{B} \cdot d\mathbf{s}$ — flux (Wb)
 $\lambda = N\Phi$ — flux linkage (Wb)

CURRENTS

$J_c = \sigma E$ — cond. density (A/m^2)
 $J_d = \epsilon \partial E / \partial t$ — disp. density
 $I_c = J_c \cdot A$ — cond. current (A)
 $I_d = J_d \cdot A$ — disp. current (A)
 $R = d/(\sigma A)$ — resistance (Ω)
 $C = \epsilon_r \epsilon_0 A/d$ — capacitance (F)
 $\tau_r = \epsilon_r \epsilon_0 / \sigma$ — relax. time (s)

MATERIALS

$\mu_0 = 4\pi \times 10^{-7}$ H/m
 μ_r — relative permeability
 $\mu = \mu_r \mu_0$ — permeability
 $\epsilon_0 = 8.85 \times 10^{-12}$ F/m
 ϵ_r — relative permittivity
 $\epsilon = \epsilon_r \epsilon_0$ — permittivity
 σ — conductivity (S/m)

CIRCUIT & WAVE

V — voltage (V)
 R — resistance (Ω)
 C — capacitance (F)
 $\omega = 2\pi f$ — angular freq (rad/s)
 $k = \omega \sqrt{\mu \epsilon}$ — wave number (m^{-1})
 $\lambda = 2\pi/k$ — wavelength (m)
 $c = 3 \times 10^8$ m/s — speed of light

PHASOR & GEOMETRY

$\eta = \sqrt{\mu/\epsilon}$ — intrinsic imp. (Ω)
 $\eta_0 = 120\pi \approx 377 \Omega$ (free space)
 $k/(\omega\mu) = 1/\eta$
 j — imaginary unit ($j^2 = -1$)
 $\tilde{\mathbf{E}}$ — phasor (drop $e^{j\omega t}$)
 A — area (m^2); d — sep. (m)
 N — turns; r — radial dist. (m)

PLANE WAVE PARAMETERS (LOSSLESS)

Angular frequency ω : rad/s; f : Hz
 $\omega = 2\pi f \quad f = \frac{\omega}{2\pi}$

Phase velocity m/s ; *nonmagnetic*: $\mu_r = 1$
 $u_p = \frac{c}{\sqrt{\epsilon_r}} \quad c = 3 \times 10^8 \text{ m/s}$

Wavenumber k rad/m
 $k = \frac{\omega}{u_p} = \frac{2\pi}{\lambda} = \frac{\omega\sqrt{\epsilon_r}}{c}$

Wavelength & impedance $m \ \& \ \Omega$
 $\lambda = \frac{u_p}{f} = \frac{2\pi}{k} \text{ (m)} \quad \eta = \frac{\eta_0}{\sqrt{\epsilon_r}} \text{ (}\Omega\text{)}, \quad \eta_0 = 377 \ \Omega$

Order: $\omega \rightarrow u_p \rightarrow k \rightarrow \lambda \rightarrow \eta$. Base: $\mathbf{E} = E_0 \cos(\omega t \pm kx + \phi_0)\hat{e}$.

LOSSLESS PLANE WAVE — E AND H

General E
 $\mathbf{E} = E_0 \cos(\omega t - k\hat{k} \cdot \mathbf{r} + \phi_0)\hat{e}$
 $\hat{k}$ = direction of travel (unit vector), e.g. $+y \Rightarrow \hat{k} = \hat{y}$.
 k = wavenumber (rad/m): how fast phase changes in space;
 $k = \omega\sqrt{\epsilon_r}/c = 2\pi/\lambda$.
 ϕ_0 = initial phase constant (rad): shift set by given condition at known t , position.
 $\hat{e}$ = polarization direction of $\mathbf{E}$ (unit vector), e.g. $\mathbf{E}$ along $x \Rightarrow \hat{e} = \hat{x}$.
 $\mathbf{r} = x\hat{x} + y\hat{y} + z\hat{z}$, so $\hat{k} \cdot \mathbf{r}$ picks the axis, e.g. $\hat{k} = \hat{y} \Rightarrow \hat{k} \cdot \mathbf{r} = y$.
 $\mathbf{H}$ from $\mathbf{E}$, $\hat{E}$, $\hat{H}$ form right-hand set
 $\mathbf{H} = \frac{1}{\eta} (\hat{k} \times \mathbf{E})$

Find ϕ_0 given $E = E_1$ at $t = 0$, position r_0
 $E_0 \cos(-kr_0 + \phi_0) = E_1 \Rightarrow \phi_0 = \cos^{-1}\left(\frac{E_1}{E_0}\right) + kr_0$
 $\cos^{-1}$ gives $[0, \pi]$; take ϕ_0 in that range.

LOSSLESS vs. LOSSY — E & H

	Lossless ($\sigma = 0$)	Lossy ($\sigma \neq 0$)
Impedance Prop.	$\eta = \sqrt{\mu/\epsilon}$ e^{-jkz}	$\eta_c = \eta_c e^{j\theta_\eta}$ $e^{-\alpha z}e^{-j\beta z}$
$\mathbf{E}(t)$	$\text{Re}\{\tilde{\mathbf{E}}e^{j\omega t}\}$	(same)
$\mathbf{H}(t)$	$\text{Re}\{\tilde{\mathbf{H}}e^{j\omega t}\}$	(same)
$\tilde{\mathbf{H}}$ from $\tilde{\mathbf{E}}$	$\frac{1}{\eta}(\hat{k} \times \tilde{\mathbf{E}})$	$\frac{1}{\eta_c}(\hat{k} \times \tilde{\mathbf{E}})$
$\tilde{\mathbf{E}}$ from $\tilde{\mathbf{H}}$	$-\eta(\hat{k} \times \tilde{\mathbf{H}})$	$-\eta_c(\hat{k} \times \tilde{\mathbf{H}})$
Phase	in phase	$\mathbf{H}$ lags $\mathbf{E}$ by θ_η

Lossless shortcuts (wave in $+z$, $\mathbf{E}$ along $\hat{e}$):
 $\mathbf{E}(z, t) = E_0 \cos(\omega t - kz + \phi)\hat{e}$
 $\mathbf{H}(z, t) = \frac{E_0}{\eta} \cos(\omega t - kz + \phi)(\hat{k} \times \hat{e})$
 $\mathbf{H}(t)$ direct $\frac{1}{\eta}(\hat{k} \times \mathbf{E}(t))$
 $\mathbf{E}(t)$ direct $-\eta(\hat{k} \times \mathbf{H}(t))$

DIRECTION OF PROPAGATION

Arg. contains	$\hat{k}$	Arg. contains	$\hat{k}$
$\omega t - kz$	$+\hat{z}$	$\omega t + kz$	$-\hat{z}$
$\omega t - ky$	$+\hat{y}$	$\omega t + ky$	$-\hat{y}$
$\omega t - kx$	$+\hat{x}$	$\omega t + kx$	$-\hat{x}$

Minus before k -axis $\Rightarrow +$ direction. Plus $\Rightarrow -$ direction.

$\hat{k} \times \hat{E}$ — FIND $\hat{H}$ DIRECTION

$\hat{k}$	$\hat{E}$	$\hat{H}$	$\hat{k}$	$\hat{E}$	$\hat{H}$
$+\hat{z}$	$\hat{x}$	$+\hat{y}$	$+\hat{z}$	$\hat{y}$	$-\hat{x}$
$-\hat{z}$	$\hat{x}$	$-\hat{y}$	$-\hat{z}$	$\hat{y}$	$+\hat{x}$
$+\hat{y}$	$\hat{x}$	$-\hat{z}$	$+\hat{y}$	$\hat{z}$	$+\hat{x}$
$-\hat{y}$	$\hat{x}$	$+\hat{z}$	$-\hat{y}$	$\hat{z}$	$-\hat{x}$
$+\hat{x}$	$\hat{y}$	$+\hat{z}$	$+\hat{x}$	$\hat{z}$	$-\hat{y}$
$-\hat{x}$	$\hat{y}$	$-\hat{z}$	$-\hat{x}$	$\hat{z}$	$+\hat{y}$

Neg. $\hat{E}$: flip $\hat{H}$ sign. $\hat{k} \times \hat{k} = \mathbf{0}$; any $\hat{a} \times \hat{a} = \mathbf{0}$.
Cyclic: $\hat{x} \times \hat{y} = \hat{z}$, $\hat{y} \times \hat{z} = \hat{x}$, $\hat{z} \times \hat{x} = \hat{y}$. Reverse $\Rightarrow$ flip sign.

TRIG PHASE SHIFTS ($\theta \equiv \omega t - kz$)

Input	Result	Input	Result
$\sin(\theta + \pi/2)$	$\cos \theta$	$\cos(\theta + \pi/2)$	$-\sin \theta$
$\sin(\theta - \pi/2)$	$-\cos \theta$	$\cos(\theta - \pi/2)$	$\sin \theta$
$\sin(\theta + \pi)$	$-\sin \theta$	$\cos(\theta + \pi)$	$-\cos \theta$
$\sin(\theta - \pi)$	$-\sin \theta$	$\cos(\theta - \pi)$	$-\cos \theta$

WAVE PARAMS

$\omega = 2\pi f$ — ang. freq (rad/s)
 $k = \omega\sqrt{\epsilon_r}/c$ — wavenumber (rad/m)
 $\lambda = 2\pi/k$ — wavelength (m)
 $u_p = c/\sqrt{\epsilon_r}$ — phase vel. (m/s)
 $\eta = \eta_0/\sqrt{\epsilon_r}$ — impedance (Ω)
 $c = 3 \times 10^8$ m/s
 $\eta_0 = 377 \ \Omega$ (free space)

LOSSLESS MEDIA

$\sigma = 0$ (vacuum, air, ideal dielectric)
 $\epsilon'' = \sigma/\omega = 0 \Rightarrow \epsilon = \epsilon'$ (real)
 $\alpha = 0$; no attenuation
 $\eta = \eta_0/\sqrt{\epsilon_r}$ real (Ω)
 $\eta_0 = 377 \ \Omega$ (free space)
 $\mathbf{E}$ and $\mathbf{H}$ in phase ($\theta_\eta = 0$)

LOSSY MEDIA

α — atten. const (Np/m)
 β — phase const (rad/m)
 $\gamma = \alpha + j\beta$ — prop. const (m⁻¹)
 η_c — complex impedance (Ω)
 $\epsilon'' = \sigma/\omega$; $\epsilon'' = \sigma/\omega$ (F/m)
 μ_r — rel. permeability ($\mu_r = 1$: nonmagnetic)
 $\mu = \mu_r\mu_0$; $\mu_0 = 4\pi \times 10^{-7}$ H/m
 $\delta_s = 1/\alpha$ — skin depth (m)

CIRCULAR WAVES

$\tilde{\mathbf{E}}_R = a_R(\hat{x} + e^{-j\pi/2}\hat{y})e^{-jkz} = a_R(\hat{x} - j\hat{y})e^{-jkz}$
 $\tilde{\mathbf{E}}_L = a_L(\hat{x} + e^{+j\pi/2}\hat{y})e^{-jkz} = a_L(\hat{x} + j\hat{y})e^{-jkz}$
 a_R, a_L, a_x in V/m; k in rad/m
Linear: $a_R = a_L = a_x/2$
 $-j = e^{-j\pi/2}$: $\cos \rightarrow \sin$
 $+j = e^{+j\pi/2}$: $\cos \rightarrow -\sin$

USEFUL NUMBERS

$\sqrt{2.56} = 1.6$; $\sqrt{9} = 3$
 $\sqrt{2.5} \approx 1.581$; $\sqrt{2} \approx 1.414$
 $\arccos(2/3) \approx 48.2^\circ = 0.84$ rad
 $\arctan(4/3) \approx 53.1^\circ = 0.927$ rad
 $\arctan(3/4) \approx 36.9^\circ = 0.644$ rad
 $\arctan(24/7) \approx 73.7^\circ = 1.287$ rad

POLARIZATION — GENERAL WAVE

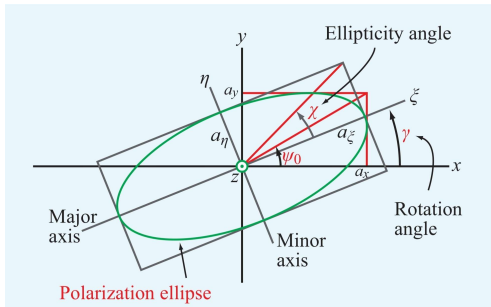


Figure 7-11 Polarization ellipse in the x - y plane with the wave traveling in the z direction (out of the page).

$\mathbf{E} = a_x \cos(\omega t - kz)\hat{x} + a_y \cos(\omega t - kz + \delta)\hat{y}$
 a_x, a_y : bounding box half-widths (V/m); δ : phase diff (rad)
 $\delta = 0$: $\mathbf{E} = (a_x\hat{x} + a_y\hat{y})\cos(\omega t - kz)$ (linear)
 $\delta = \pi$: $\mathbf{E} = (a_x\hat{x} - a_y\hat{y})\cos(\omega t - kz)$ (linear)
Auxiliary angle $\psi_0 = \arctan(a_y/a_x) \in (0, \pi/2)$
Rotation angle γ (major axis from $\hat{x}$) $\gamma \in [0, \pi)$
 $\tan(2\gamma) = \tan(2\psi_0) \cos \delta$
If $\text{sign}(\gamma) \neq \text{sign}(\delta)$: $\gamma \leftarrow \gamma \pm \frac{\pi}{2}$.
Ellipticity angle χ $\chi \in [-\pi/4, \pi/4]$
 $\sin(2\chi) = \sin(2\psi_0) \sin \delta$
Axial ratio $R = 1/|\tan \chi|$ major axis = $R \times$ minor; $R \geq 1$

χ SHAPE & HANDEDNESS

χ describes the shape of the ellipse and rotation direction.

χ (rad)	Polarization Type	Handedness
0	Linear	None
$(0, \pi/4)$	RHEP	RH (CCW)
$\pi/4$	RHC	RH (CCW)
$-\pi/4$	LHC	LH (CW)
$(-\pi/4, 0)$	LHEP	LH (CW)

QUADRANT RULE FOR $\tan(2\gamma)$

$N = \sin(2\psi_0) \cos \delta, \quad D = \cos(2\psi_0)$			
N	D	2γ (deg)	2γ (rad)
+	+	$\arctan(N/D)$	$\arctan(N/D)$
+	-	$180^\circ - \arctan N/D $	$\pi - \arctan N/D $
-	-	$180^\circ + \arctan N/D $	$\pi + \arctan N/D $
-	+	$360^\circ - \arctan N/D $	$2\pi - \arctan N/D $

$\gamma = (2\gamma)/2$. $D = 0 \Rightarrow 2\gamma = \frac{\pi}{2} \Rightarrow \gamma = \frac{\pi}{4}$.
Convert: $\theta_{\text{rad}} = \theta_0 \times \frac{\pi}{180}$; $\theta_0 = \theta_{\text{rad}} \times \frac{180}{\pi}$.

PHASOR $\leftrightarrow$ TIME DOMAIN

Phasor coeff	$\text{Re}\{\cdot\} e^{j\omega t} e^{-\alpha z} e^{-j\beta z}$
A ($= Ae^{j\phi}$)	$A \cos(\omega t - \beta z)$
$-jA$ ($= Ae^{-j\pi/2}$)	$A \sin(\omega t - \beta z)$
$+jA$ ($= Ae^{+j\pi/2}$)	$-A \sin(\omega t - \beta z)$
$-A$ ($= Ae^{\pm j\pi}$)	$-A \cos(\omega t - \beta z)$
$Ae^{j\phi}$ (general)	$A \cos(\omega t - \beta z + \phi)$
$A + jB$ ($= Z e^{j\theta}$)	$ Z \cos(\omega t - \beta z + \theta)$

$|Z| = \sqrt{A^2 + B^2}$, $\theta = \arctan(B/A)$ (check z quadrant)

CIRCULAR POLARIZATION — TIME DOMAIN

$a_x = a_y = a$, wave in $+\hat{z}$
RHC $\delta = -\pi/2$; $\tilde{\mathbf{E}} = a(\hat{x} - j\hat{y})e^{-jkz}$
 $\mathbf{E}(z, t) = a[\cos(\omega t - kz)\hat{x} + \sin(\omega t - kz)\hat{y}]$
LHC $\delta = +\pi/2$; $\tilde{\mathbf{E}} = a(\hat{x} + j\hat{y})e^{-jkz}$
 $\mathbf{E}(z, t) = a[\cos(\omega t - kz)\hat{x} - \sin(\omega t - kz)\hat{y}]$

LOSSY MEDIA — CLASSIFICATION

Loss tangent

ϵ''/ϵ'	Type	Use
> 100	Good conductor	GC approx
0.01 to 100	Quasi-conductor	Exact eqs
< 0.01	Low-loss dielectric	LD approx

$\epsilon' = \epsilon_r \epsilon_0$; $\epsilon'' = \sigma/\omega$; nonmag.: $\mu = \mu_0$; free space: $\epsilon_r = \mu_r = 1$.

ANY MEDIUM — EXACT ($X = \sqrt{1 + (\epsilon''/\epsilon')^2}$)

$\alpha = \omega \sqrt{\frac{\mu\epsilon'}{2}} \sqrt{X-1}$ Np/m, $\beta = \omega \sqrt{\frac{\mu\epsilon'}{2}} \sqrt{X+1}$ rad/m
 $\lambda = \frac{2\pi}{\beta} = \frac{u_p}{f}$ (m), $u_p = \frac{\omega}{\beta}$ (m/s)
 $\eta_c = \frac{\eta}{X^{1/2}} e^{j\theta_\eta}$ (Ω), $\theta_\eta = \frac{1}{2} \arctan \frac{\sigma}{\omega\epsilon'}$ (rad)
Rect: $\eta_c = |\eta_c| \cos \theta_\eta + j|\eta_c| \sin \theta_\eta$, $|\eta_c| = \eta/X^{1/2}$
 $\eta = \eta_0/\sqrt{\epsilon_r}$; $\epsilon' = \epsilon_r \epsilon_0$.

SKIN DEPTH δ_s (m) — ANY LOSSY MEDIUM

General: $\delta_s = 1/\alpha$ $|\mathbf{E}(z)| = E_0 e^{-z/\delta_s}$; lossless: $\alpha = 0 \Rightarrow$ no decay
Low-loss ($\epsilon''/\epsilon' \ll 1$) $\delta_s \approx 2/(\sigma\eta)$
Good conductor ($\epsilon''/\epsilon' \gg 1$) $\delta_s \approx 1/\sqrt{\pi f \mu \sigma}$

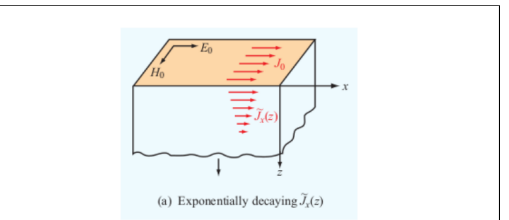
LOW-LOSS MEDIUM ($\epsilon''/\epsilon' \ll 1$)

$\alpha \approx \frac{\sigma}{2} \sqrt{\frac{\mu}{\epsilon'}} = \frac{\sigma\eta}{2}$ (Np/m)
 $\beta \approx \omega \sqrt{\mu\epsilon'} = \frac{\omega\sqrt{\epsilon_r}}{c}$ (rad/m), $u_p \approx \frac{c}{\sqrt{\epsilon_r}}$ (m/s)
 $\eta_c \approx \sqrt{\frac{\mu}{\epsilon'}} = \eta$ (Ω , $\angle \approx 0$)

GOOD CONDUCTOR ($\epsilon''/\epsilon' \gg 1$)

$\alpha \approx \beta \approx \sqrt{\pi f \mu \sigma}$ (Np/m, rad/m)
 $u_p \approx \sqrt{4\pi f / \mu \sigma}$ (m/s), $\lambda \approx u_p/f$ (m)
 $\eta_c \approx (1+j)\sqrt{\frac{\pi f \mu}{\sigma}} \approx (1+j)\frac{\alpha}{\sigma}$ (Ω)

SURFACE RESISTANCE (GOOD CONDUCTOR)



$J_0 = \sigma E_0$ — current density at surface ($z=0$)
 $\tilde{\mathbf{J}}(z) = J_0 e^{-(1+j)z/\delta_s} \hat{x}$ — decays with depth
 $R_s = \sqrt{\pi f \mu / \sigma} = 1/(\sigma \delta_s)$ (Ω)
 $Z_s = (1+j)R_s$ — surface impedance
 $R_{AC} = R_s \ell/w$ — AC resistance of slab

AVERAGE POWER DENSITY (POYNTING)

General W/m^2
 $S_{av} = \frac{1}{2} \text{Re}\{\tilde{\mathbf{E}} \times \tilde{\mathbf{H}}^*\}$
Lossless shortcut $|\mathbf{E}_0|^2 = a_x^2 + a_y^2 + a_z^2$
 $S_{av} = \frac{|\mathbf{E}_0|^2}{2\eta} \hat{k}$
Lossy medium $\eta_c = |\eta_c| e^{j\theta_\eta}$; wave in $+\hat{z}$
 $S_{av} = \frac{|\tilde{\mathbf{E}}(0)|^2}{2|\eta_c|} e^{-2\alpha z} \cos \theta_\eta \hat{z}$
Direction = $\hat{k}$. If wave goes in $-\hat{x}$, then S_{av} is a negative x -component.

DECIBELS & ATTENUATION

$G = 10 \log(P_1/P_2) = 20 \log(|E_1|/|E_2|)$ (dB)
 $A = -8.69 \alpha z$ (dB); α (dB/m) $\equiv 8.69 \alpha$ (Np/m)